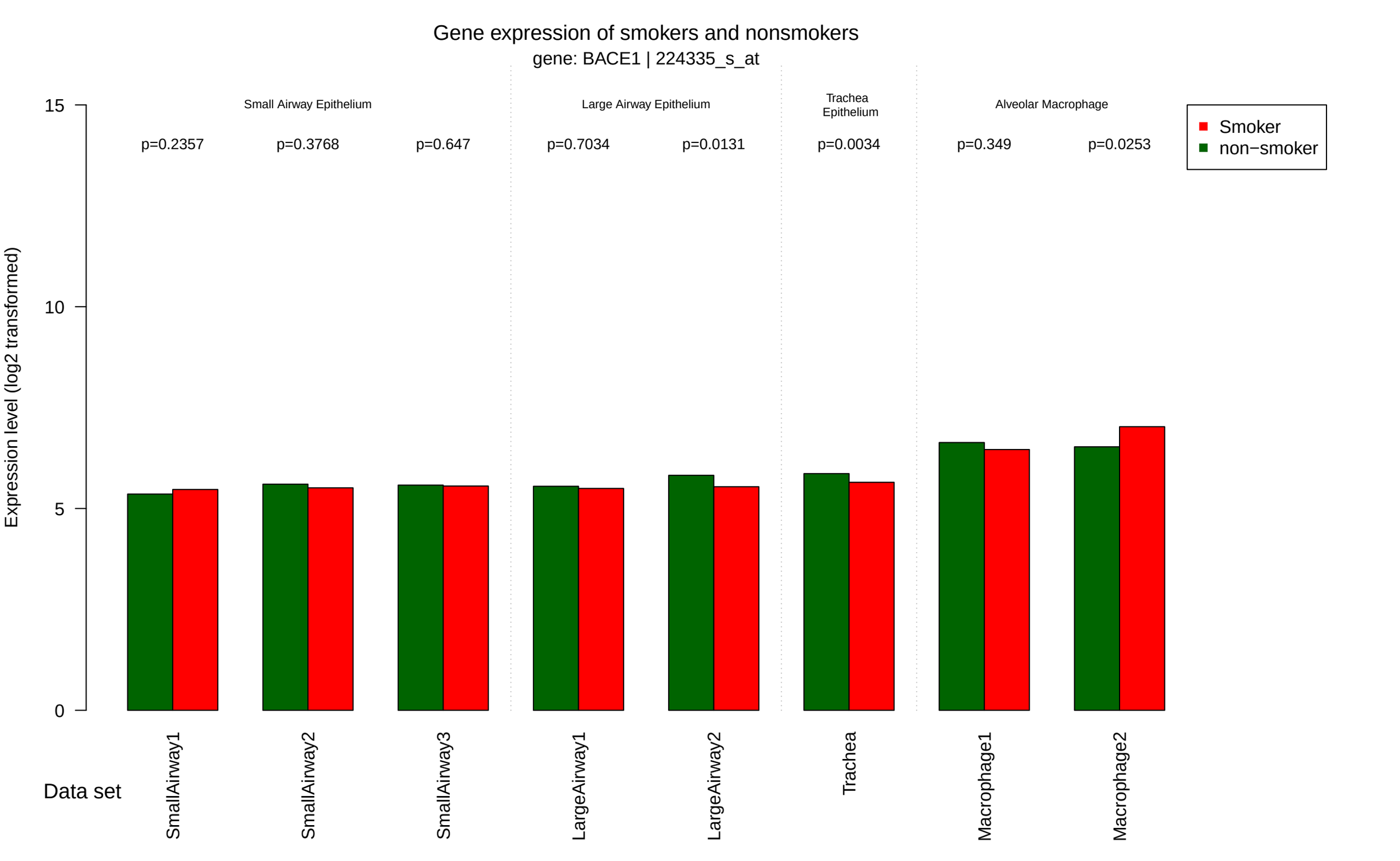
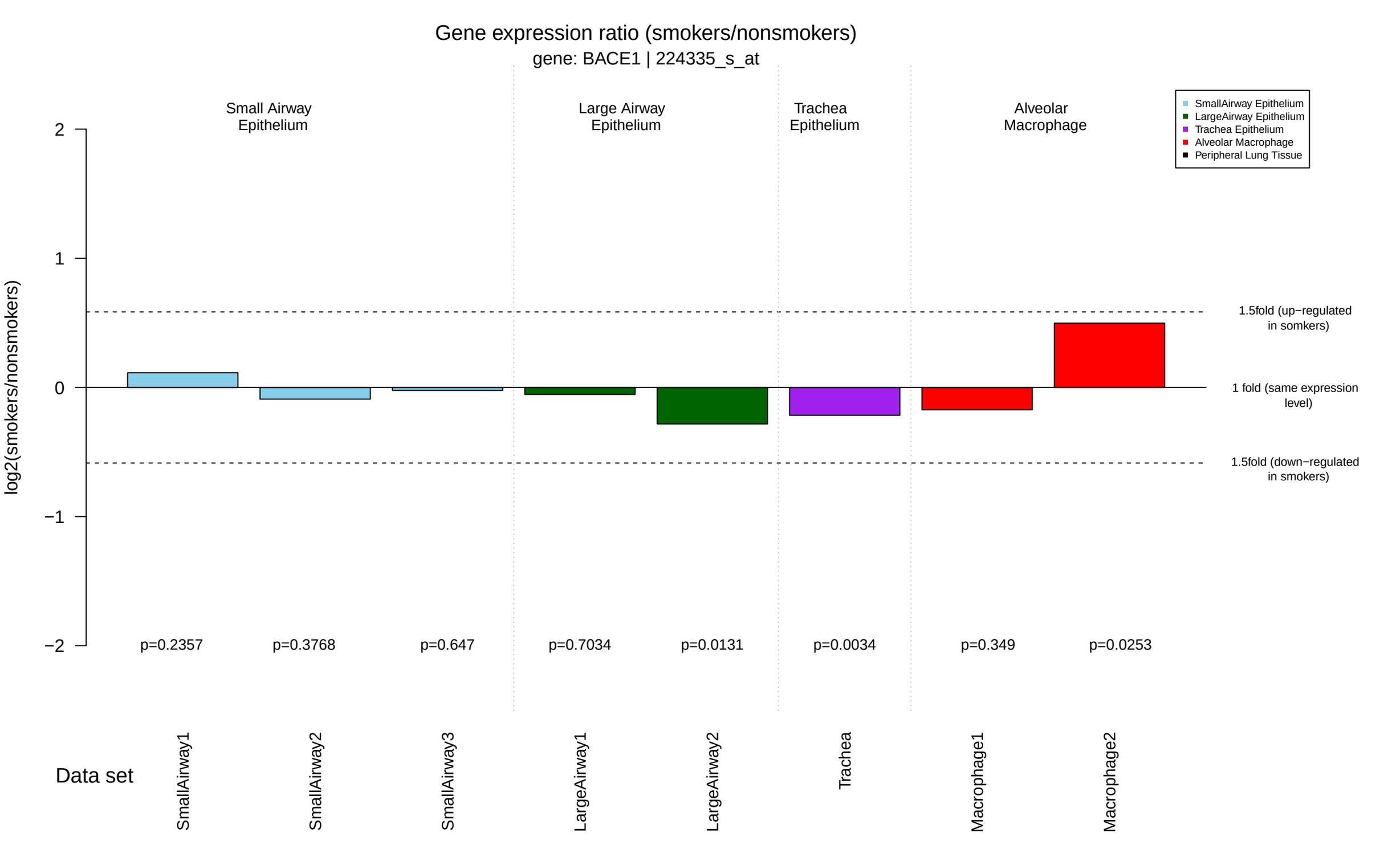
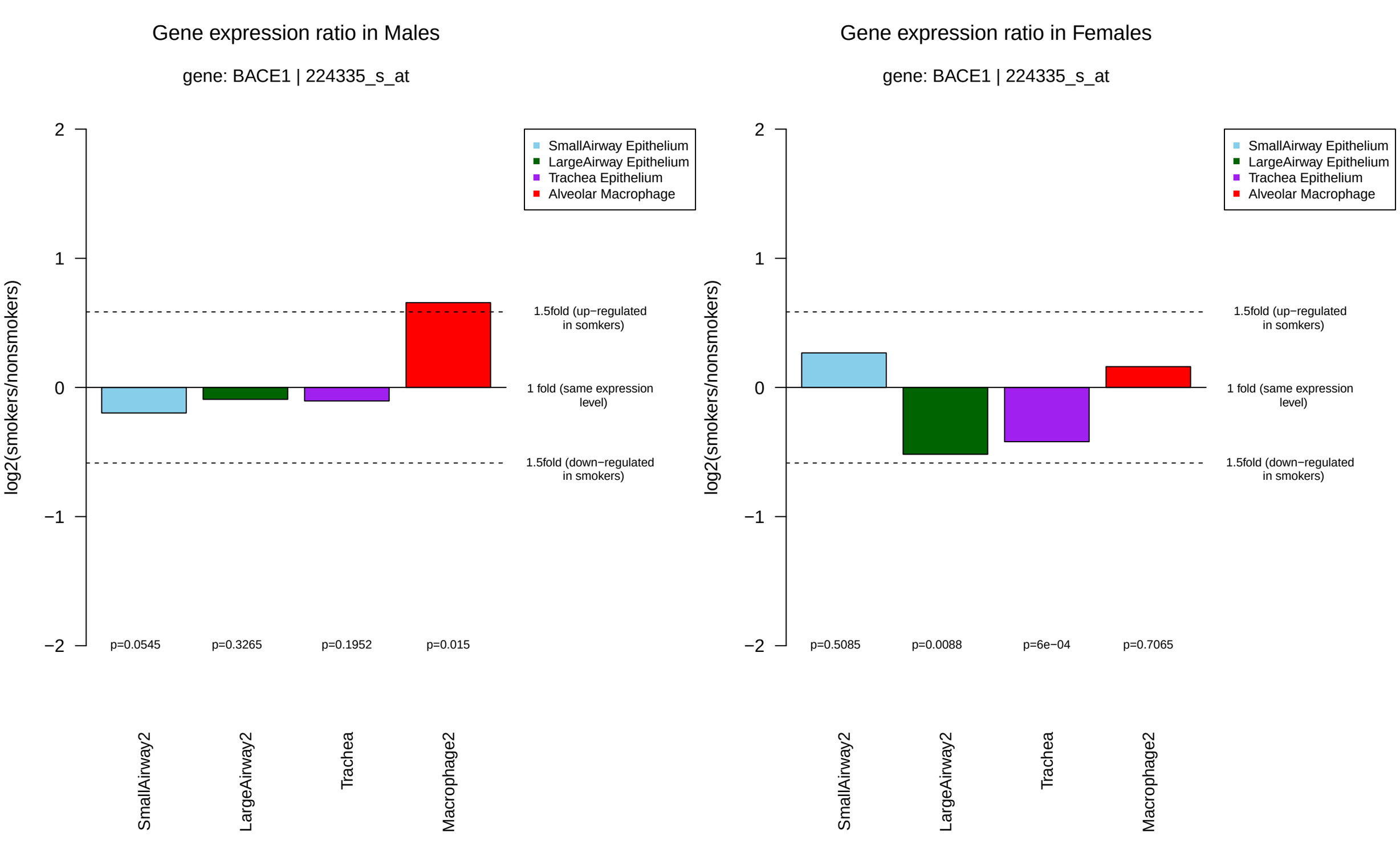
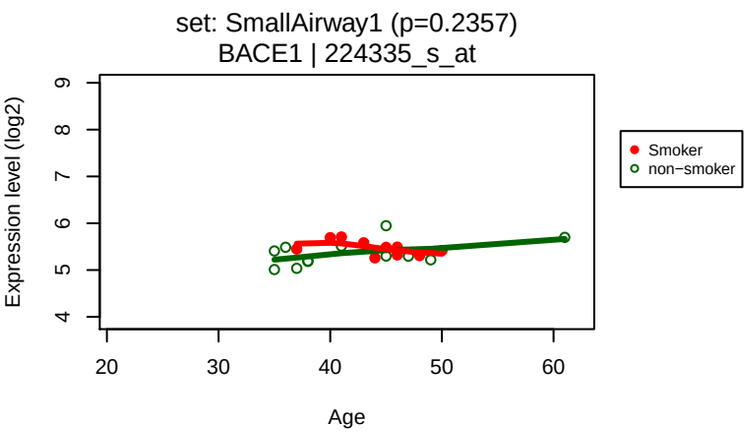
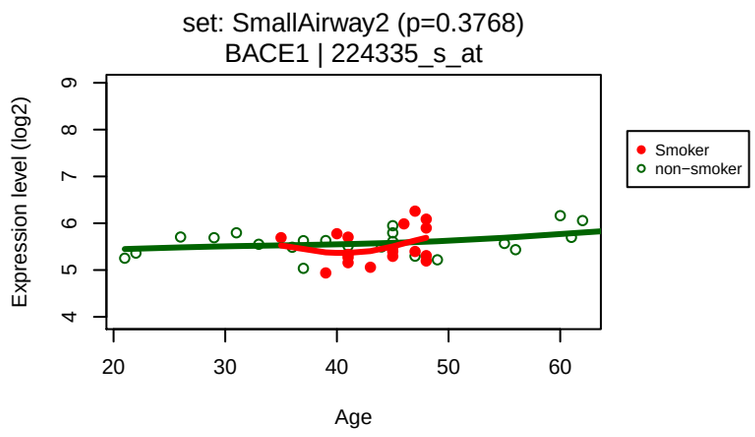
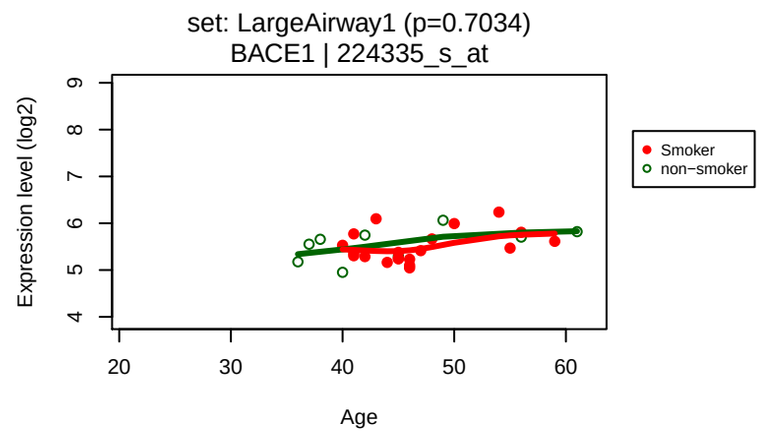

Gene expression of smokers and nonsmokers
gene: BACE1 | 224335_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2357
p=0.3768
p=0.647
p=0.7034
p=0.0131
p=0.0034
p=0.349
p=0.0253
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

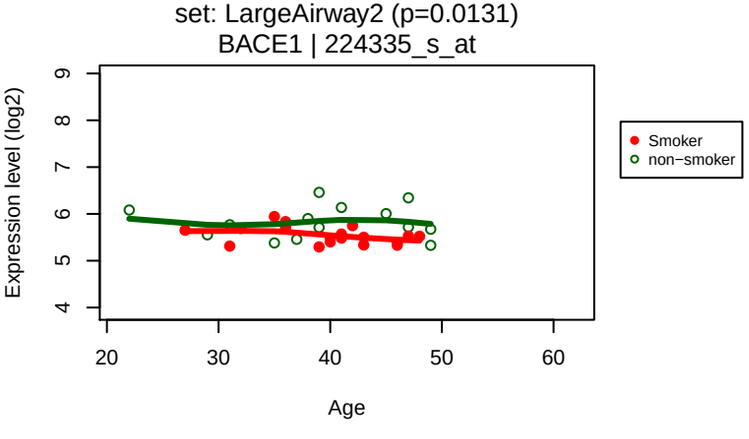

Gene expression ratio (smokers/nonsmokers)
gene: BACE1 | 224335_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2357
p=0.3768
p=0.647
p=0.7034
p=0.0131
p=0.0034
p=0.349
p=0.0253
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

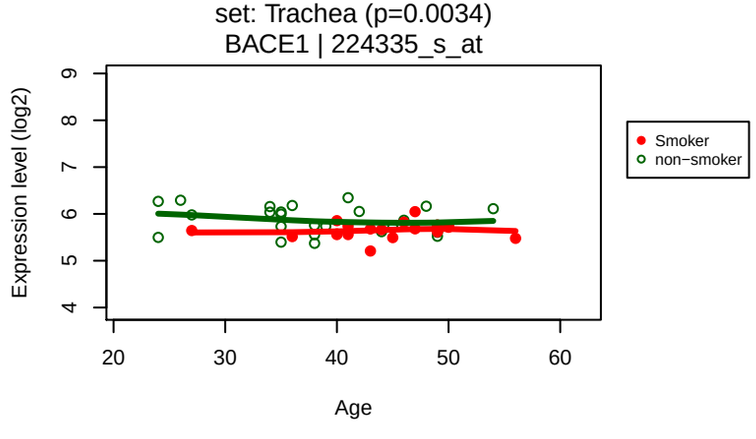

Gene expression ratio in Males
gene: BACE1 | 224335_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0545
p=0.3265
p=0.1952
p=0.015
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BACE1 | 224335_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5085
p=0.0088
p=6e−04
p=0.7065
SmallAirway2
LargeAirway2
Trachea
Macrophage2

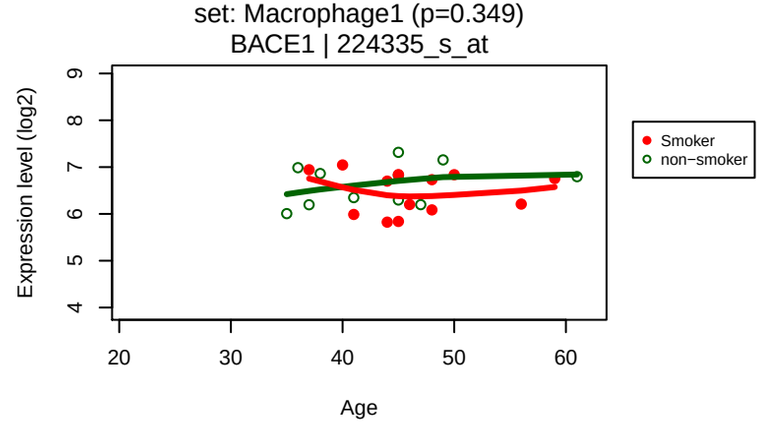

set: SmallAirway1 (p=0.2357)
BACE1 | 224335_s_at
Expression level (log2)
Age
Smoker
non−smoker

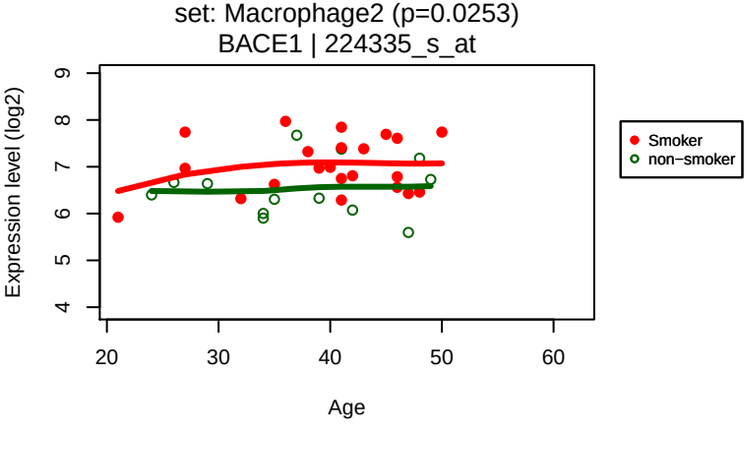

set: SmallAirway2 (p=0.3768)
BACE1 | 224335_s_at
Expression level (log2)
Age
Smoker
non−smoker

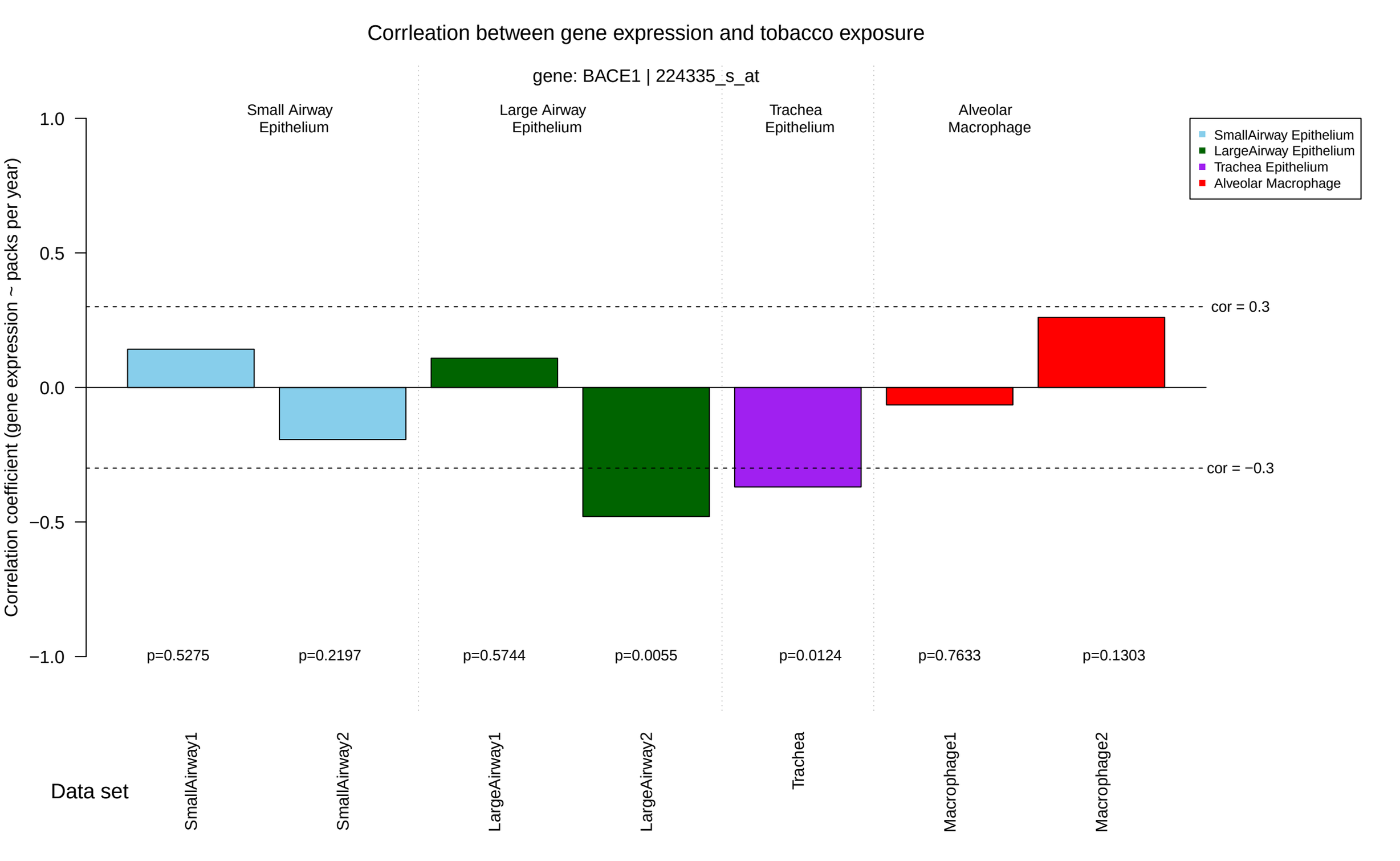

set: LargeAirway1 (p=0.7034)
BACE1 | 224335_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0131)
BACE1 | 224335_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0034)
BACE1 | 224335_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.349)
BACE1 | 224335_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0253)
BACE1 | 224335_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BACE1 | 224335_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5275
p=0.2197
p=0.5744
p=0.0055
p=0.0124
p=0.7633
p=0.1303
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2